SEMINAR SUMMARY: FEBRUARY 10, 2026 - DAVID CHAN HUMAN/INDIVIDUAL RESEARCH

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The goal behind the talk by Professor Chan on February 10$^{\text{th}}$ was to explore three new research topics that we did not go over in his initial talk on January 12$^{\text{th}}$. More specifically we covered the research topics: *Contact Tracing*, *Student Support Networks*, and *Instructor Student Interaction*. We will explore the different research projects below in the subsequent sections.

1. Contact Tracing

Contact-tracing is the physical process of retracing human interconnectivity over a brief window of time in hopes of determining a full graph of distinct people that a single individual had contact with during that window of time to help determine and mitigate potential disease spread. During the Covid pandemic we saw remedial efforts at contact tracing in the population, but because the disease's symptoms were so mild during the period of contagion for an individual and the symptoms of the disease were so mild for a considerable portion of the population we were largely unable to do anything about it via these processes. Where we see hugely effective contact tracing is in populations where the disease is far more lethal and the mechanism of transfer between patients is non-airbourne and more based in physical contact. For example in Africa, the infrastructure and knowledge around contact tracing due to Ebola is profoundly better than anywhere else in the world, and it remains the bed of knowle

The project compares a Watts-Strogatz Network model with the standard ordinary differential equation model. The Watts-Strogatz Network model is a probabilistic mechanism for developing edges on an order N graph. The input variables for the edge generation algorithm (which we will omit as it is widely available in the literature) are N as the number of nodes in the graph, K is the average degree, and β is a rough probability of connection (note we have intentionally been imprecise). An example of a graph generated in this fashion follows as:

Our model we then put through a series of states as given in Figure 2. Furthermore, the model assumes that symptomatic individuals get tested at some rate σ then get quarantined, and for individuals in quarantine, their contacts are tested at a rate τ.

Date: February 13, 2026.

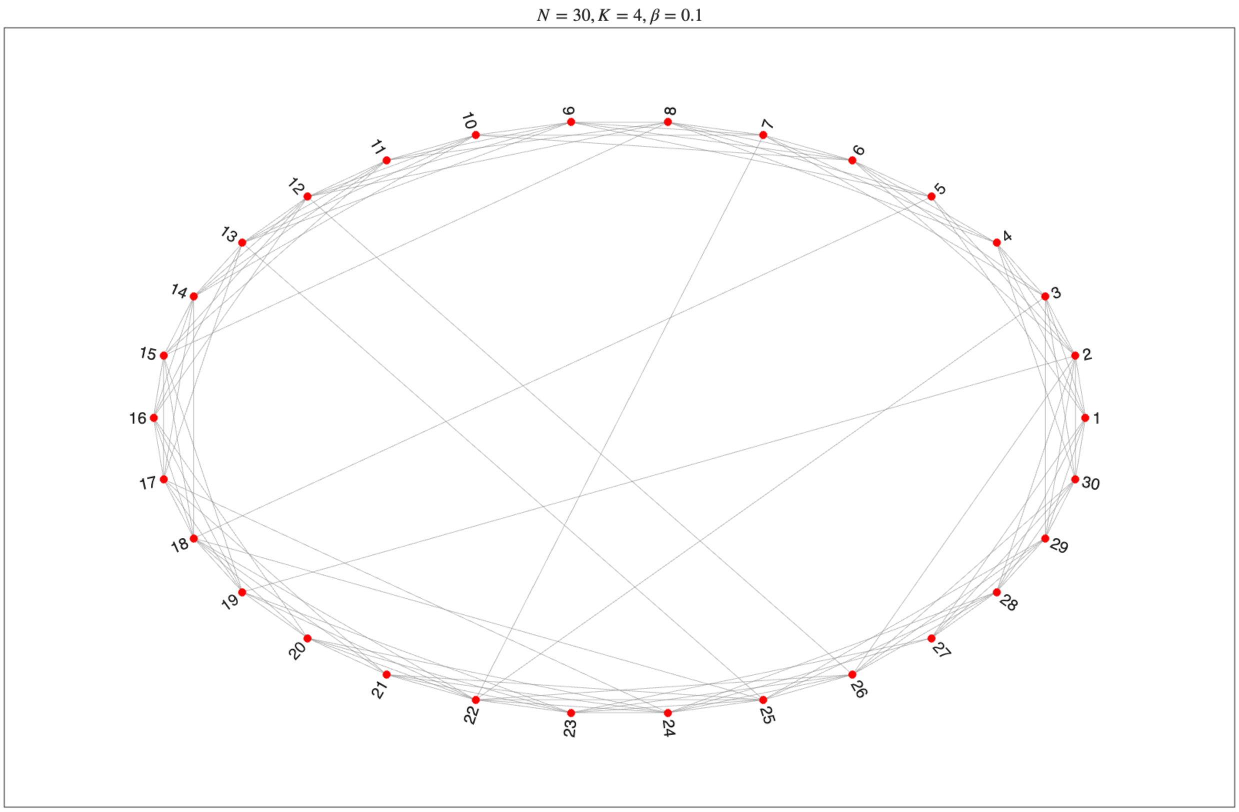

Figure 1. A Watts-Strogats graph

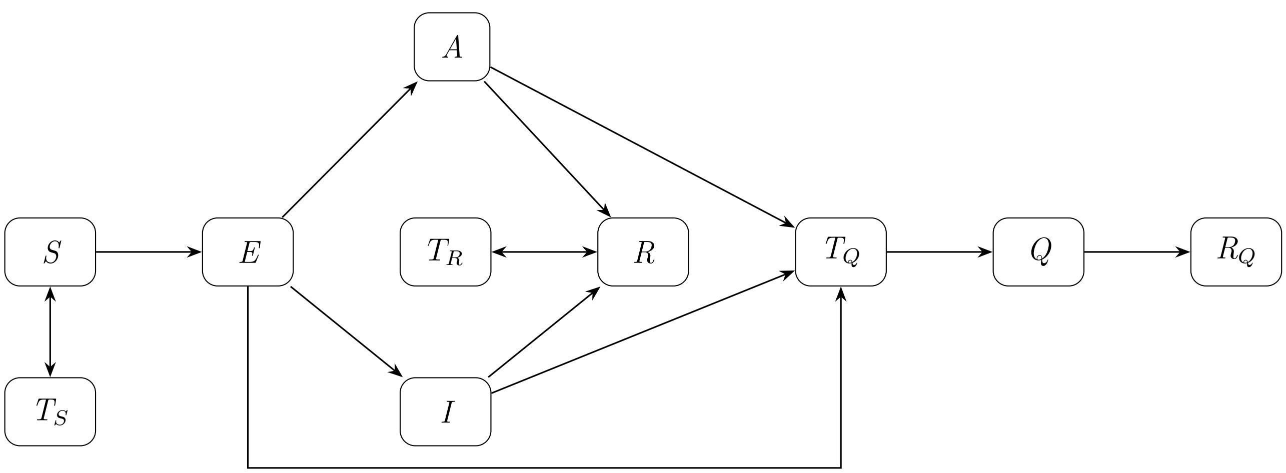

Figure 2. Model Diagram

In Figure 2, each variable represents a population at a stage in the process of infection. For a population X, T_X represents the individuals of population X that test for the disease. The populations in the diagram follow as S-the susceptible, E-the exposed, A-the asymptomatic, I-the infected, R-the recovered, and Q-the quarantined.

Experimentation. The main question that explored was "how often did an outbreak occur?" We looked at heap plots with $K = 20$ and $K = 40$ over (τ, σ), and what we found was that that if people frequently self-identify, and they test well then we don't see disease outbreaks. On the other hand, if people don't self-identify and don't test then we have disease outbreaks. The ODE model for this case of frequency of outbreak did quite poorly because it assumes a complete graph on our nodes as opposed to the Watts-Strogatz model. In fact, it predicted that there would be an outbreak in every case regardless of whether there was testing and self-identifying.

Instead of considering just how often did an outbreak occur, some other variables were considered for comparing models namely: length of the epidemic, the number of waves the epidemic had, considering if everyone was infected, and considering if everyone tested regularly. When we looked at the length of the epidemic ad the number of waves plots, the gradients for the ODE models particularly stood out as useful. Though, when everyone was infected the models behaved in a similar fashion to the original outlook of frequency of outbreaks, the ODE model went back to having close to a singleton for the entire heap. Interestingly, when testing was involved, the dynamic of the epidemic changed in regard to

Learnings. The first point realized was the type of model quite matters. Though, that's merely for us mathematicians. What we really noticed though was that self-reporting (and education) matters, effective accurate timely testing matters, and social distancing matters. So apparently those lessons we learned as a global community from Ebola translate to airbourne diseases as well.

2. Student Support Networks

This project seemed to be a psychology study. It looked like there was some valuable data and good lessons to learn.

3. Instructor Student Interaction

I suppose that it does make sense that we would have a wealth of data about MATH200 classes, and that we could use some linear difference equations. That said, the data that I'm most interested in personally is the student data from BIOL 300 (Cellular & Mollecular Biology) mainly because it has a reputation from outside of class of being quite a rich set of data that spans the Covid pandemic. I had not learned until class that the data spans dempgraphics, grades, seating, and technology use.